

Dietary factors and incident atrial fibrillation: the Framingham Heart Study



Background: There have been conflicting reported associations between dietary factors and incident atrial fibrillation (AF). Objective: We evaluated associations between consumption of alcohol, caffeine, fiber, and polyunsaturated fatty acids (PUFAs) and incident AF in the Framingham Heart Study. Design: Participants without AF (n = 4526; 9640 examinations; mean age: 62 y; 56% women) from the original and offspring cohorts completed food-frequency questionnaires and were followed prospectively for 4 y. We examined the associations between dietary exposures and AF with Cox proportional hazards regression. Results: A total of 296 individuals developed AF (177 men, 119 women). In multivariable analyses, there were no significant associations between examined dietary exposures and AF risk. Hazard ratios (HRs) for increasing quartiles of dietary factors were as follows: for alcohol, 0.73 (95% CI: 0.5, 1.05), 0.85 (95% CI: 0.61, 1.18), and 1.12 (95% CI: 0.83, 1.51) (P for trend = 0.48); for caffeine, 0.84 (95% CI: 0.62, 1.15), 0.87 (95% CI: 0.64, 1.2), and 0.98 (95% CI: 0.7, 1.39) (P for trend = 0.84); for total fiber, 0.86 (95% CI: 0.61, 1.2), 0.64 (95% CI: 0.44, 0.92), and 0.81 (95% CI: 0.54, 1.2) (P for trend = 0.16); and for n-3 (omega-3) PUFAs, 1.11 (95% CI: 0.81, 1.54), 0.92 (95% CI: 0.65, 1.29), and 1.18 (95% CI: 0.85, 1.64) (P for trend = 0.57; quartile 1 was the reference group). In exploratory analyses, consumption of >4 servings of dark fish/wk (5 cases and 21 individuals at risk) was significantly associated with AF risk compared with the consumption of <1 serving of dark fish/wk (HR: 6.53; 95% CI: 2.65, 16.06; P < 0.0001). Conclusions: Consumption of alcohol, caffeine, fiber, and fish-derived PUFAs was not significantly associated with AF risk. The observed adverse association between the consumption of dark fish and AF merits further investigation. Our findings suggest that the dietary exposures examined convey limited attributable risk of AF in the general population.